

Control traffic flow and security to my VPC

Carrying on from my previous project where I created my VPC.

I will create a security group inside my subnet. What is a security group you may ask well a security group is a security guard at the entrance for each building in the subnet for each resource. Security groups are responsible for checking who can enter or leave the resource by checking the IP address and protocols (are different ways to move data, like HTTP ,FTP) and Port Numbers (where the data will enter or exist, each port is designed for a specific protocol). While creating my security groups I learnt the difference between inbound and outbound, which is inbound means to control the data that can enter my resources and outbound means to control the data that can leave the resource.

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
HTTP	TCP	80	An... 0.0.0.0/0	
Add rule				

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type Info	Protocol	Port range Info	Destination Info	Description - optional Info
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Security group (sg-0a81ca8f71e65c489 | NextWork Security Group) was created successfully
[Details](#)

sg-0a81ca8f71e65c489 - NextWork Security Group [Actions](#)

Details

Security group name NextWork Security Group	Security group ID sg-0a81ca8f71e65c489	Description A Security Group for the NextWork VPC.	VPC ID vpc-0ccdc714e6f8be733
Owner 360529135867	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Regarding the outbound rules AWS security groups already allow outbound traffic, unless I specify otherwise. Now since we are done with the route table and basic security groups sorted for my VPC. To level my VPC I need to add Network ACL. The network ACL does what a security group does for its resources but the difference is that the network ACL does it for the whole subnet, for example they can block incoming traffic from a range of IP addresses or deny access to certain ports. Where a security group can specify which ports and protocols are allowed for each resource.

Filter inbound rules

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Rule number	Type	Protocol	Port range	Source	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

So with custom Network ACL's AWS denys traffic unless we edit the rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	✔ Allow
*	All traffic	All	All	0.0.0.0/0	⊗ Deny

I have associated my network ACL with a subnet